

Context

Tourism is now recognized as a directly measurable activity, enabling more accurate analysis and more effective policies can be made for tourism. Whereas previously the sector relied mostly on approximations from related areas of measurement (e.g. Balance of Payments statistics), tourism nowadays is a productive activity that can be analyzed using factors like economic indicators, social indicators, environmental & infrastructure indicators, etc. As a Data Scientist in a leading tours and travels company, you have been assigned the task of analyzing several of these factors and group countries based on them to help understand the key locations where the company can invest in tourism services.

Objective

To explore the data and identify different groups of countries based on important factors to find key locations where investments can be made to promote tourism services.

Key Questions

1. Which variables should be used for clustering?
2. How many different groups/clusters of countries can be found from the data?
3. How do the different clusters vary?
4. How to use PCA to retain the components which explain 90% variance?
5. How to perform clustering using the components obtained from PCA?

Data Description

This dataset contains key statistical indicators of the countries. It covers sections like general information, economic indicators, social indicators, environmental & infrastructural indicators.

Data Dictionary

- country: country
- Region: region of the country
- Surface area (km²): Surface area in sq. km
- Population in thousands (2017): Population of the country, in thousands, as in the year 2017
- Population density (per km², 2017): Population density per km², as in the year 2017

- Sex ratio (m per 100 f, 2017): Number of males per 100 female, as in the year 2017
- GDP: Gross domestic product (million current US\\\$): GDP of the country in million USD
- Economy: Agriculture (% of GVA): Contribution of agriculture to the economy as a percentage of Gross Value Added
- Economy: Industry (% of GVA): Contribution of the industry to the economy as a percentage of Gross Value Added
- Economy: Services and other activity (% of GVA): Contribution of services and other activities to the economy as a percentage of Gross Value Added
- International trade: Exports (million US dollar): Amount, in million USD, of international exports
- International trade: Imports (million US dollar): Amount, in million USD, of international imports
- International trade: Balance (million US dollar): Amount, in million USD, of balance between international exports and imports
- Fertility rate, total (live births per woman): Fertility rate of the country computed as the no. of live births per woman
- Infant mortality rate (per 1000 live births): Infant mortality rate of the country computed as the no. of dead infants per 1000 live births
- Health: Total expenditure (% of GDP): Total expenditure on healthcare facilities as a percentage of GDP
- Education: Government expenditure (% of GDP): Total expenditure on education as a percentage of GDP
- Mobile-cellular subscriptions (per 100 inhabitants): no. of mobile/cellular subscriptions per 100 people
- Individuals using the Internet (per 100 inhabitants): no. of individuals using the Internet per 100 people
- Threatened species (number): Number of threatened species
- CO2 emission estimates (million tons/tons per capita): CO2 emission estimates in million tons
- Energy production, primary (Petajoules): energy production in petajoules

Learning Outcomes

- Exploratory Data Analysis
- Clustering
- Cluster Profiling
- Dimensionality Reduction
- Business Recommendations

Steps and Tasks

- Import libraries and load dataset
- EDA

- Perform clustering and cluster profiling
- Perform dimensionality reduction
- Perform clustering and cluster profiling on lower-dimensional data
- Provide business recommendations and conclusion